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Week 1: Fundamentals of Speech Signals, Sampling, LTI Systems & Spectral Analysis

> **Track:** Speech Technology (IIT Madras BS Degree — BSEE4001)
> **Scope:** Week 1 of 12
> **Core Focus:** Continuous acoustic production, Nyquist-Shannon sampling, amplitude quantization, LTI discrete convolution, rational all-pole resonators, the Discrete Fourier Transform, autocorrelation pitch tracking, STFT spectrograms, source-filter cepstral separation, and Mel filter bank acoustic feature extraction.

Abstract

This document constitutes the comprehensive technical and mathematical reference for Week 1 of Speech Technology. We examine the complete acoustic signal processing pipeline, bridging

continuous-time physical acoustics to discrete representations suitable for statistical modeling and neural acoustic feature extraction.

We formalize:

1. The five-stage **Speech Chain** connecting cognitive linguistic intent to auditory decoding.
2. The continuous-to-discrete conversion boundary governed by the **Nyquist-Shannon Sampling Theorem** ($f_s \geq 2f_{max}$), folding aliasing dynamics, and uniform amplitude quantization noise ($\text{SQNR} \approx 6.02B + 1.76$ dB).
3. The **four-transform Fourier matrix** (CTFS, CTFT, DTFS, DTFT), formalizing the **Discrete Fourier Transform (DFT)** and its Fast Fourier Transform (FFT) algorithmic implementation as an orthogonal coordinate projection.
4. **Linear Time-Invariant (LTI) Systems**, difference equations, the discrete convolution sum ($y[n] = x[n] * h[n]$), and rational transfer functions $H(z)$ whose complex conjugate pole pairs define vocal tract formant resonances.
5. Statistical **autocorrelation** for fundamental frequency (F_0) pitch estimation and Voiced/Unvoiced (V/UV) classification.
6. The **Short-Time Fourier Transform (STFT)**, analyzing the time-frequency trade-offs between **Wideband** and **Narrowband spectrograms**.
7. Gunnar Fant's **Source-Filter Model** ($S = E \cdot V \cdot R$), separated via **homomorphic cepstral deconvolution** in the quefrency domain.
8. Psychoacoustic **Mel-scale filter bank** construction mirroring human cochlear frequency integration.

All concepts are accompanied by publication-grade scientific figures and verifiable Python implementations.

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1. The Speech Chain & Acoustic Foundations

1.1 The Five-Stage Communication Pipeline

Speech communication transforms an abstract neural thought in the speaker's brain into a decoded linguistic message in the listener's brain across five sequential physical and physiological domains:

\begin{aligned} \text{Speaker Brain (Linguistic)} &\xrightarrow{\text{Motor Neurons}} \text{Vocal Articulators (Physiological)} \\ &\xrightarrow{\text{Air Compression}} \text{Pressure Wave } x(t) \text{ (Acoustic)} \\ &\xrightarrow{\text{Eardrum Vibration}} \text{Cochlea/Basilar Membrane (Physiological)} \\ &\xrightarrow{\text{Auditory Nerve}} \text{Listener Brain (Linguistic)} \end{aligned}

- **Automatic Speech Recognition (ASR):** Maps the physical acoustic pressure wave into the linguistic sequence of words/phonemes.
- **Text-to-Speech (TTS):** Synthesizes linguistic text into an acoustic pressure wave.

1.2 Acoustic Units: Phonemes, Pitch, and Formants

1. **Phonemes:** The atomic distinctive linguistic units of sound. Categorized into:
   - **Vowels:** Unobstructed vocal tract shapes characterized by distinct resonant peaks (/a/, /i/, /u/).
   - **Consonants:** Produced with vocal tract constrictions or closures, classified by *manner* (stops /p, t, k/, fricatives /s, f/, nasals /m, n/, liquids /l, r/, glides /w, j/) and *place* (bilabial, alveolar, velar).
2. **Voiced vs. Unvoiced Sound Generation:**
   - **Voiced Sounds:** Produced by aerodynamic self-oscillation of the vocal folds within the larynx. Generates a quasi-periodic train of airflow volume pulses characterized by fundamental frequency F_0 (pitch).
   - **Unvoiced Sounds:** Produced with open (abducted) vocal folds. Turbulent acoustic noise is generated by forcing air through narrow constrictions (e.g., /s/, /sh/).
3. **Pitch (F_0):** The rate of vocal fold vibration. Adult males typically range from 80\text{--}150\ \mathrm{Hz}; adult females from 150\text{--}250\ \mathrm{Hz}; children exceed $300\ \mathrm{Hz}$. Pitch conveys prosody, emphasis, and emotion, but does not alter vowel identity.
4. **Formants (F_1, F_2, F_3):** The natural acoustic resonant frequencies of the vocal tract cavity. Formant peaks determine vowel identity:
   - F_1 is inversely correlated with tongue height: high vowels (/i/, /u/) exhibit low F_1 ($300\ \mathrm{Hz}$); low vowels (/a/) exhibit high F_1 ($800\ \mathrm{Hz}$).
   - F_2 is positively correlated with tongue advancement: front vowels (/i/) exhibit high F_2 ($2200\ \mathrm{Hz}$); back vowels (/u/) exhibit low F_2 ($800\ \mathrm{Hz}$).

1.3 Psychoacoustics & Human Auditory Perception

- **Non-Linear Frequency Sensitivity:** Pitch perception is linear below $1000\ \mathrm{Hz}$ and logarithmic above $1000\ \mathrm{Hz}$.
- **Auditory Masking:** A loud acoustic sound renders a quieter sound in adjacent frequency bins (**spectral masking**) or adjacent time intervals (**temporal masking**) inaudible.
- **Equal-Loudness (Fletcher-Munson Contours):** Human hearing is most sensitive between $2\ \mathrm{kHz}$ and $5\ \mathrm{kHz}$ (ear canal resonance) and falls off sharply below $100\ \mathrm{Hz}$ and above $10\ \mathrm{kHz}$.

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2. Continuous-to-Discrete Foundations: Sampling & Quantization

2.1 The Nyquist-Shannon Sampling Theorem

A continuous-time analog speech signal $x(t)$ with continuous Fourier transform $X(\Omega)$ is sampled uniformly at intervals $T_s = 1/f_s$. Modeling the sampled signal $x_s(t)$ as multiplication by an ideal Dirac impulse comb:

$$x_s(t) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s)$$

Applying the Continuous-Time Fourier Transform (F) yields:

$$X_s(\Omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(\Omega - k\Omega_s), \text{where } \Omega_s = 2\pi f_s$$

The spectrum of the discrete sequence contains infinitely replicated copies of the analog spectrum spaced at integer multiples of f_s.

$$\text{Nyquist-Shannon Criterion: } f_s \geq 2f_{\max}$$

If $f_s \geq 2f_{\max}$, an ideal low-pass filter with cutoff $\Omega_c = \Omega_s/2$ perfectly reconstructs $x(t)$ via the Whittaker-Shannon interpolation formula:

x(t) = \sum_{n=-\infty}^{\infty} x[n] \operatorname{sinc}\left(\frac{t - n T_s}{T_s}\right), \quad \operatorname{sinc}(\theta) = \frac{\sin(\pi \theta)}{\pi \theta}

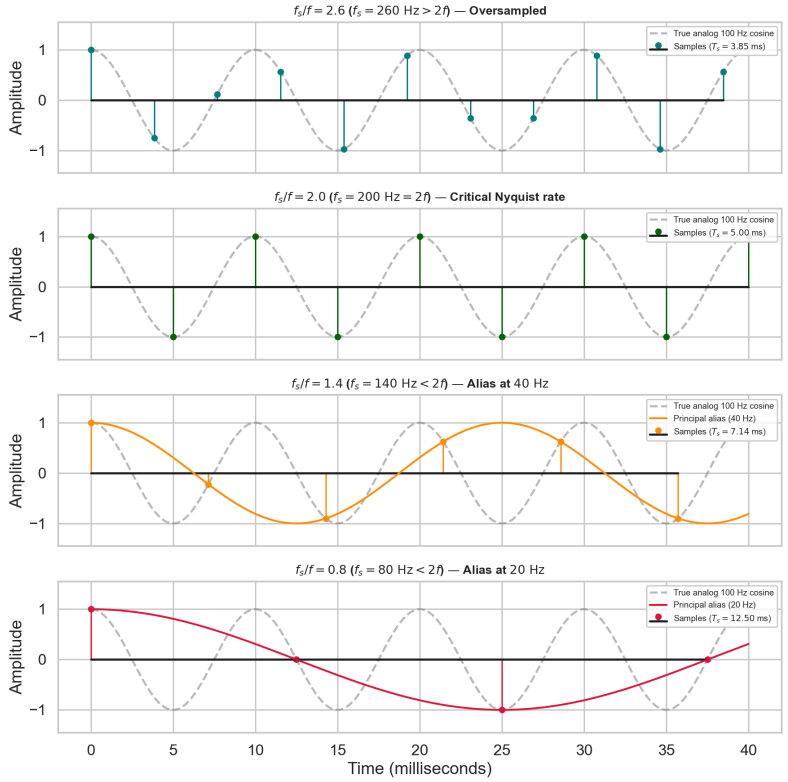


Figure 1. *Nyquist-Shannon sampling and aliasing dynamics. Demonstrating reconstruction fidelity across sampling ratios* $f_s/f = 2.6$ *(oversampled),* 2.0 *(critical Nyquist limit),* 1.4 *(undersampled alias at* 40 Hz*), and* 0.8 *(severe undersampling alias at* 20 Hz*). Analog reference:* 100 Hz *sinusoid.*

2.2 Aliasing Distortion & Folding Formula

When a continuous sinusoid $x(t) = \cos(2\pi f t)$ is sampled at $f_s < 2f$, the apparent perceived digital frequency f_{apparent} folds into the fundamental Nyquist interval $[0, f_s/2]$ according to:

f_{\mathrm{apparent}} = \left| f - k \cdot f_s \right|, \quad \text{where } k = \operatorname{round}\left(\frac{f}{f_s}\right)

To eliminate aliasing, all hardware Analog-to-Digital Converters (ADCs) place an analog **anti-aliasing low-pass filter** ($f_c \leq f_s/2$) prior to sampling.

2.3 Amplitude Quantization & The $6.02\ \mathrm{dB}$ Rule

Quantization maps continuous sample amplitudes into 2^B discrete levels. For a uniform quantizer over range $[-X_{\max}, +X_{\max}]$, the step size is $\Delta = \frac{2X_{\max}}{2^B}$. Under the standard assumption of uniformly distributed quantization noise $e[n] \sim \mathrm{U}[-\Delta/2, +\Delta/2]$, the noise power is:

$$P_e = \sigma_e^2 = \frac{\Delta^2}{12} = \frac{X_{\max}^2}{3 \cdot 2^{2B}}$$

For a full-scale sinusoid ($P_x = X_{\max}^2/2$), the theoretical Signal-to-Quantization-Noise Ratio is:

$$\mathrm{SQNR} = 10\log_{10}\left(\frac{P_x}{P_e}\right) = 10\log_{10}\left(\frac{3}{2} \cdot 2^{2B}\right) \approx 6.02B + 1.76\ \mathrm{dB}$$

Every added bit of quantization precision increases dynamic range by $6.02\ \mathrm{dB}$. A 16-bit linear PCM system provides $\approx 98\ \mathrm{dB}$ of dynamic range, sufficient to encompass the full dynamic range of human hearing without audible quantization hiss.

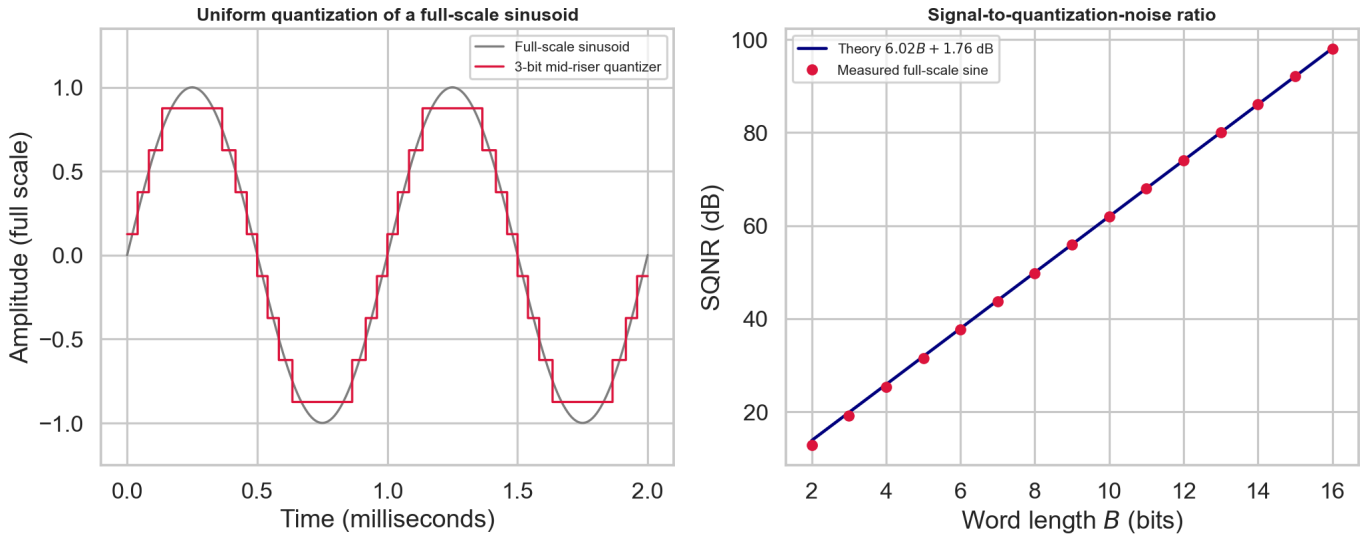


Figure 2. *Uniform quantization mapping and the theoretical 6.02 dB per bit SQNR limit against measured error on a full-scale sinusoid.*

3. Signal Transforms & The Discrete Fourier Transform Engine

3.1 The Fourier Transform Taxonomy

Transform	Time Domain Input	Frequency Domain Output	Mathematical Formulation
CTFS (Continuous Fourier Series)	Continuous & Periodic (T_0)	Discrete & Aperiodic ($k\Omega_0$)	$c_n = \frac{1}{T_0}\int_{T_0} x(t)e^{-jn\Omega_0 t}dt$
CTFT (Continuous Fourier Transform)	Continuous & Aperiodic	Continuous & Aperiodic (Ω)	$X(\Omega)=\int_{-\infty}^{\infty} x(t)e^{-j\Omega t}dt$
DTFS (Discrete Fourier Series)	Discrete & Periodic (N)	Discrete & Periodic (N)	$a_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}$
DTFT (Discrete-Time Fourier Transform)	Discrete & Aperiodic	Continuous & Periodic (2π)	$X(\omega)=\sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$
DFT (Discrete Fourier Transform)	**Discrete & Finite (N)**	**Discrete & Periodic (N)**	$X[k]=\sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}$

3.2 Basis Function Representation in Hilbert Space

Any vector or signal in a linear inner-product space can be decomposed as a linear combination of orthogonal basis vectors:

$$x[n]=\sum_{k=0}^{N-1} c_k \phi_k[n], c_k = \frac{\langle x,\phi_k \rangle}{\|\phi_k\|_2}$$

For the DFT, the orthogonal basis sequences are discrete complex sinusoids: $\phi_k[n]=e^{j\frac{2\pi}{N}kn}$.

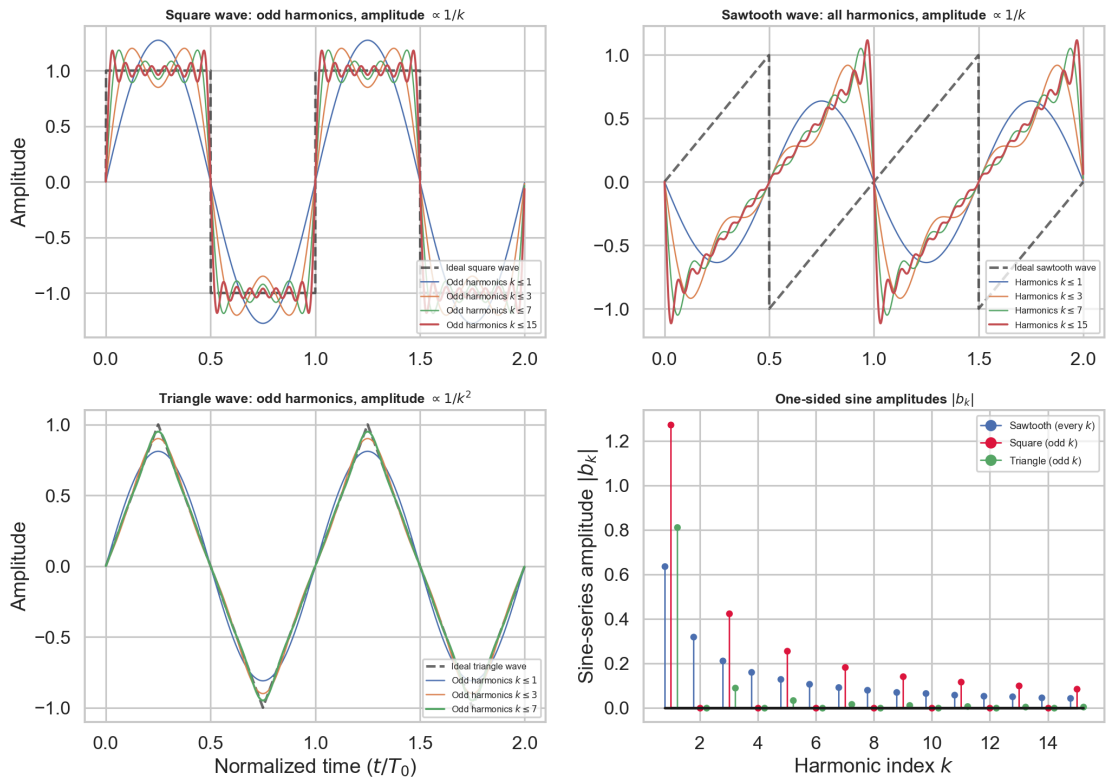


Figure 3. *Harmonic Fourier series decomposition of periodic waveforms. Square wave (odd harmonics, $1/k$ decay), Sawtooth wave (all harmonics, $1/k$ decay), and Triangle wave (odd harmonics, rapid $1/k^2$ decay). Right panel shows discrete line spectra $|c_k|$ as a function of harmonic index k.*

3.3 The Forward and Inverse DFT

Forward DFT (Analysis):

X[k] = \sum_{n=0}^{N-1} x[n] W_N^{k n}, \quad k = 0, 1, \dots, N-1 \quad \left(W_N = e^{-j \frac{2\pi}{N}}\right)

Inverse DFT (Synthesis):

x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-k n}, \quad n = 0, 1, \dots, N-1

3.4 Frequency Bin Resolution & Real-Signal Symmetry

Each discrete frequency index k maps to physical continuous frequency f_k according to:

$$f_k = k \cdot \frac{f_s}{N} = k \cdot \Delta f \,(\text{Hz})$$

The **frequency resolution** is strictly:

$$\Delta f = \frac{f_s}{N} = \frac{1}{NT} = \frac{1}{T_{\text{record}}}$$

For real speech sequences ($x[n] \in \mathbb{R}$), the DFT obeys **conjugate symmetry**:

X[N - k] = X^*[k] \implies |X[N - k]| = |X[k]|, \quad \angle X[N - k] = -\angle X[k]
All unique spectral information resides within bins k = 0, 1, \dots, N/2, spanning frequencies from $0\ \mathrm{Hz}$ to the Nyquist folding frequency $f_s/2$.

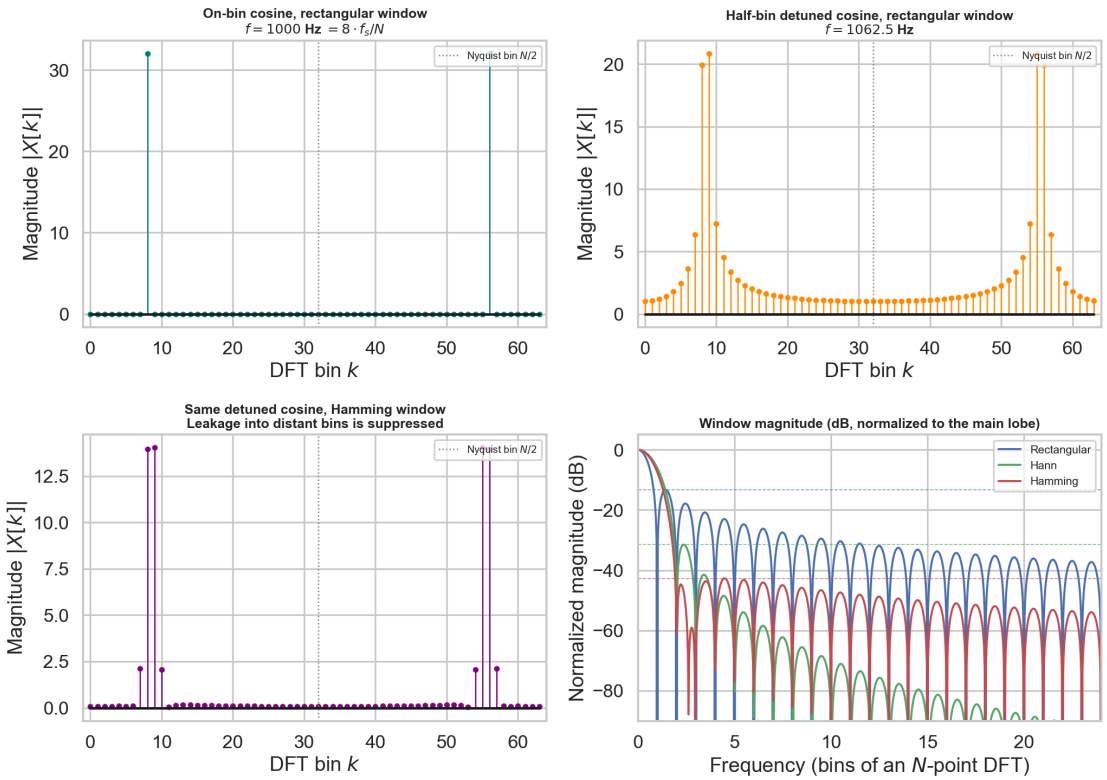


Figure 4. *Spectral leakage and window function smoothing. Top: Rectangular vs. Hamming window on bin-centered and off-bin frequencies. Bottom: Sidelobe decay profiles (dB) for Rectangular, Hann, and Hamming windows.*

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4. Linear Time-Invariant Systems & Rational Digital Filters

4.1 System Property Definitions & Proofs

A discrete system is an operator $\mathrm{T}\{\cdot\}$ transforming input $x[n]$ to output $y[n]$.

1. **Linearity (Superposition):**

$$\mathrm{T}\{ax_1[n]+bx_2[n]\}=a\mathrm{T}\{x_1[n]\}+b\mathrm{T}\{x_2[n]\}$$

2. **Time-Invariance (Shift-Invariance):**

$$\text{If } \mathrm{T}\{x[n]\}=y[n], \text{then } \mathrm{T}\{x[n-k]\}=y[n-k] \forall k \in \mathbb{Z}$$

3. **Causality:** Output $y[n]$ depends only on present and past inputs \{x[n], x[n-1], x[n-2], \dots\}.
4. **BIBO Stability:** Every bounded input ($|x[n]| \le B_x < \infty$) yields a bounded output ($|y[n]| \le B_y < \infty$).

Detailed Proof: System $y[n]=nx[n]$

- *Linearity:* $\mathrm{T}\{ax_1[n]+bx_2[n]\}=n(ax_1[n]+bx_2[n])=a(nx_1[n])+b(nx_2[n])=ay_1[n]+by_2[n]$. **LINEAR.**
- *Time-Invariance:* Delayed input gives $y_1[n]=nx[n-k]$. Delayed output gives $y[n-k]=(n-k)x[n-k]$. Because $n\neq n-k$, **TIME-VARIANT**.
- *Causality & Memory:* Depends only on current index n. **CAUSAL & MEMORYLESS**.

Detailed Proof: System $y[n]=e^{x[n]}$

- *Linearity:* $\mathrm{T}\{ax_1[n]+bx_2[n]\}=e^{ax_1[n]+bx_2[n]}=(e^{x_1[n]})^a(e^{x_2[n]})^b\neq ae^{x_1[n]}+be^{x_2[n]}$. **NON-LINEAR.**
- *Time-Invariance:* Delayed input gives $y_1[n]=e^{x[n-k]}$. Delayed output gives $y[n-k]=e^{x[n-k]}$. Because $y_1[n]\equiv y[n-k]$, **TIME-INVARIANT**.
- *Causality & Memory:* Depends only on current index n. **CAUSAL & MEMORYLESS**.

4.2 Discrete Convolution & The Convolution Theorem

For an LTI system with impulse response $h[n]=\mathrm{T}\{\delta[n]\}$:

$$y[n]=x[n]*h[n]=\sum_{k=-\infty}^{\infty}x[k]h[n-k]$$

By the Convolution Theorem:

$$y[n] = x[n] * h[n] \quad \stackrel{\mathcal{F}}{\longleftrightarrow} \quad Y(\omega) = X(\omega) \cdot H(\omega)$$

Time-domain convolution corresponds to point-wise complex multiplication in frequency.

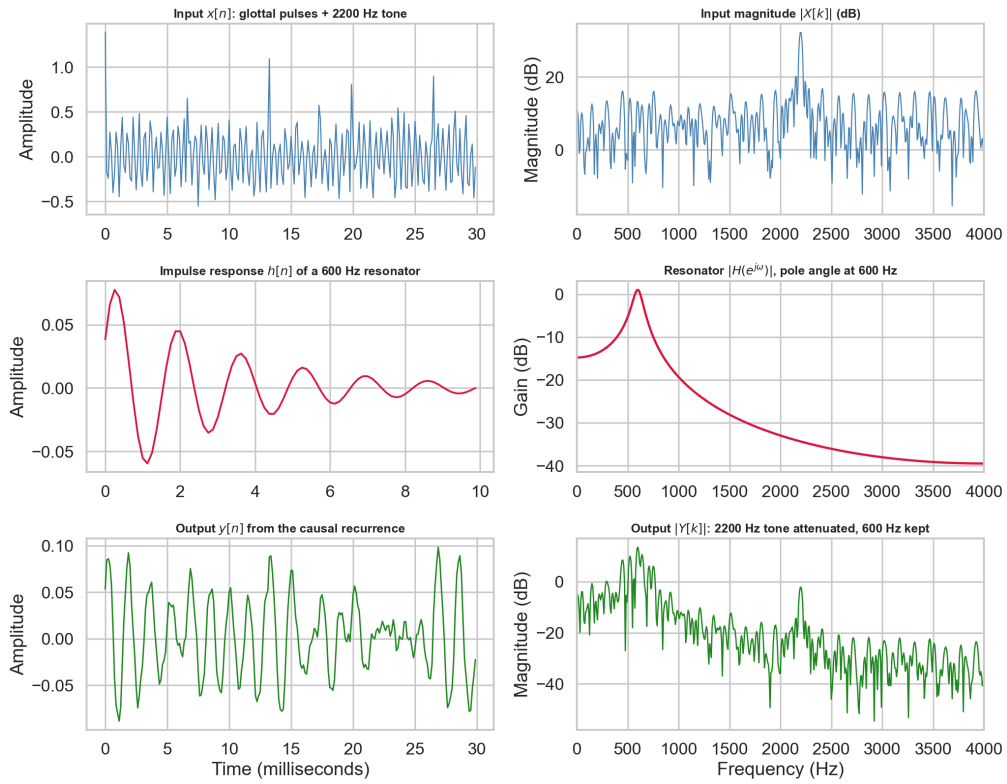


Figure 5. *LTI system discrete convolution and spectral filtering. Top row: Input sequence* $x[n]$ *containing glottal pulses contaminated by high-frequency noise (*$2200\ \text{Hz}$*) and its magnitude spectrum. Middle row: Vocal tract impulse response* $h[n]$ *and resonant filter transfer function* $|H(\omega)|$ *(*$F_1 = 600\ \text{Hz}$*). Bottom row: Output speech sequence* $y[n] = x[n] * h[n]$ *and output spectrum* $|Y(\omega)| = |X(\omega)| \cdot |H(\omega)|$*, demonstrating selective formant amplification.*

4.3 Rational Filters, Poles & Zeros in the Z-Plane

A digital filter is governed by the linear constant-coefficient difference equation:

$$y[n] - \sum_{k=1}^{N} a_k\, y[n-k] = \sum_{r=0}^{M} b_r\, x[n-r]$$

Taking the Z-transform yields the rational system transfer function $H(z)$:

$$H(z) = \frac{\sum_{r=0}^{M} b_r\, z^{-r}}{1 - \sum_{k=1}^{N} a_k\, z^{-k}} = K \frac{\prod_{i=1}^{M} (z - z_i)}{\prod_{k=1}^{N} (z - p_k)}$$

- **Zeros (z_i):** Roots of numerator; transmission nulls ($H(z_i) = 0$).
- **Poles (p_k):** Roots of denominator; resonant peaks ($H(p_k) \to \infty$).
- **Causality & Stability:** A causal LTI system is BIBO stable if and only if **all poles lie strictly inside the unit circle** ($|p_k| < 1$).

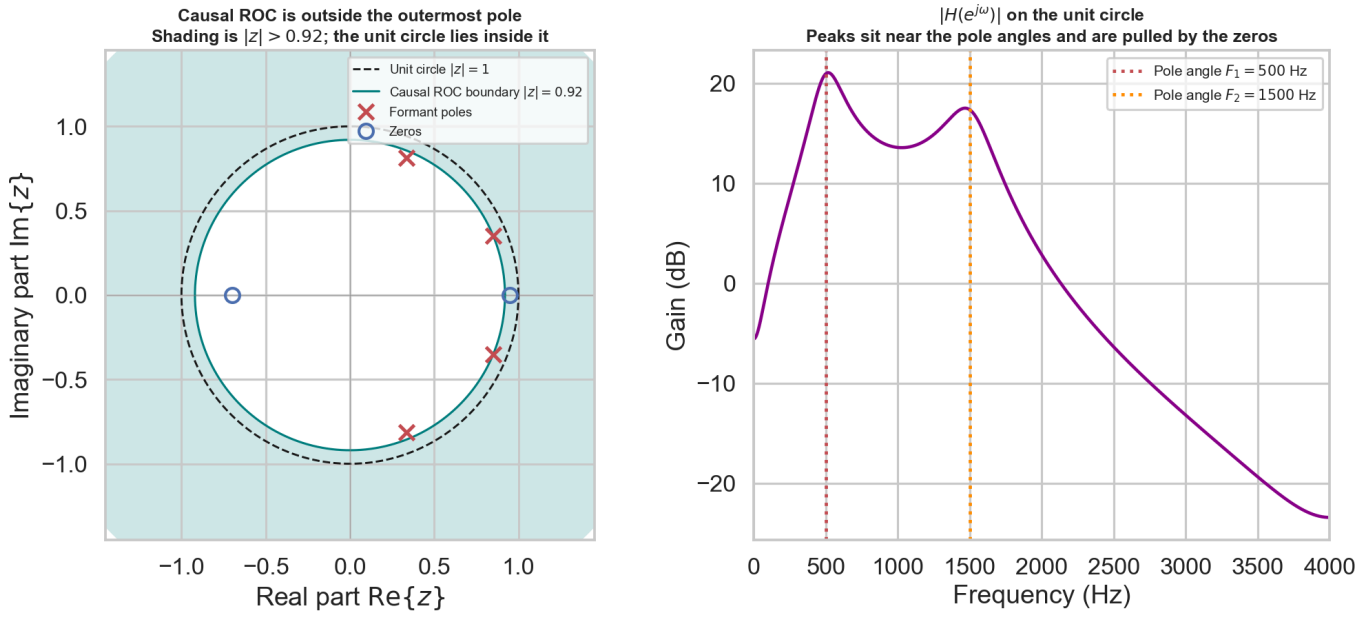


Figure 6. *Complex Z-plane pole-zero constellation and frequency response. Left: Resonant formant poles (×) located at conjugate angles inside the unit circle $|z|=1$, with shaded ROC $|z|>\max|p_k|$ ensuring stability. Right: Magnitude response along the unit circle $|H(e^{j\omega})|$ displaying formant resonance peaks at $F_1=500$ Hz and $F_2=1500$ Hz.*

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5. Statistical Correlation & Pitch (F_0) Tracking

5.1 The Autocorrelation Sequence

For a real discrete speech sequence $x[n]$, the deterministic autocorrelation sequence is:

$$r_{xx}[l]=\sum_{n=-\infty}^{\infty} x[n]x[n-l]$$

The normalized autocorrelation coefficient is $\varrho_{xx}[l]=\frac{r_{xx}[l]}{r_{xx}[0]}$.

Core Theorems:

1. **Zero-Lag Energy Maximum:** $r_{xx}[0]=\sum_n x^2[n]=E_x$.
2. **Even Symmetry:** $r_{xx}[l]=r_{xx}[-l]$.
3. **Cauchy-Schwarz Upper Bound:** $|r_{xx}[l]|\leq r_{xx}[0]$ for all $l\in\mathbb{Z}$.
4. **Periodicity Preservation:** If $x[n]=x[n+N_0]$, then $r_{xx}[l]=r_{xx}[l+N_0]$.

5.2 Pitch Tracking & Voiced/Unvoiced (V/UV) Classification

In voiced speech, vocal fold pulses recur every $N_0 = f_s / F_0$ samples, producing a dominant peak at lag $l_* = N_0$. We restrict search to physiological limits ($50\ \text{Hz} \le F_0 \le 500\ \text{Hz}$):

$$L_{\min} = \lfloor \frac{f_s}{500} \rfloor, L_{\max} = \lceil \frac{f_s}{50} \rceil$$

l^* = \arg\max_{l \in [L_{\min}, L_{\max}]} \rho_{xx}[l] \implies F_0 = \frac{f_s}{l^*}

$$\text{Decision Rule: Frame is Voiced if } \varrho_{xx}[l_*] \ge 0.30, \text{Unvoiced otherwise}$$

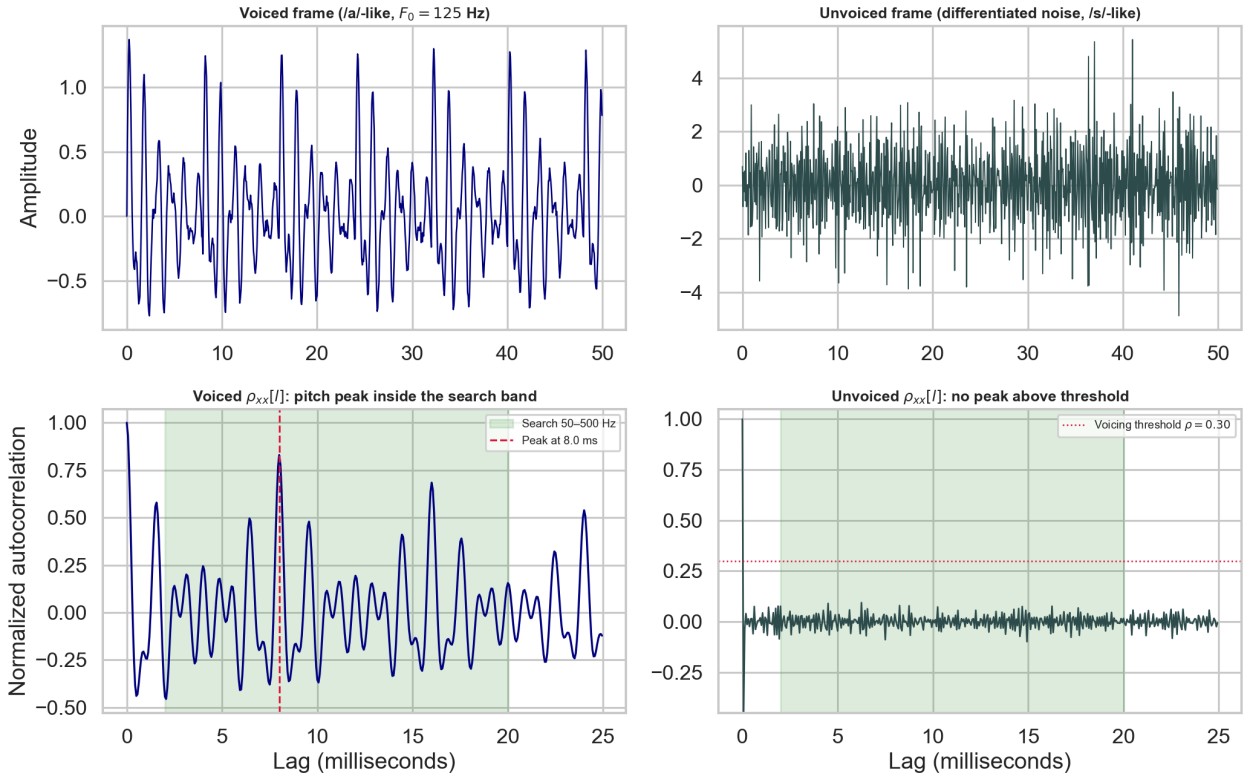


Figure 7. *Normalized autocorrelation functions for voiced speech (/a/, left) and unvoiced speech (/s/, right). Voiced speech displays a prominent pitch peak at lag $l_* = 128$ samples (8.0 ms, $F_0 = 125$ Hz). Unvoiced speech decays rapidly to zero without secondary periodic maxima.*

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6. Non-Stationary Speech Processing: STFT & Spectrograms

6.1 The Short-Time Fourier Transform (STFT)

Because speech is non-stationary, it is analyzed over short, quasi-stationary frames (20\text{--}30\ \mathrm{ms}) shifted by R samples (10 ms) using the STFT:

$$X(m,\omega) = \sum_{n=-\infty}^{\infty} x[n] w[n - mR] e^{-j\omega n}$$

The **spectrogram** is the squared magnitude: $S(m,\omega) = |X(m,\omega)|_2$.

6.2 The Wideband vs. Narrowband Spectrogram Trade-Off

Governed by the Gabor-Heisenberg limit $\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$:

- **Wideband Spectrogram (L \approx 5\text{--}10\ \mathrm{ms}):** High time resolution (Δt small); resolves individual vocal cord closures as **vertical pitch striations** and broad formant bands.
- **Narrowband Spectrogram (L \approx 40\text{--}100\ \mathrm{ms}):** High frequency resolution (Δf small); resolves discrete pitch harmonics ($k \cdot F_0$) as **horizontal lines**.

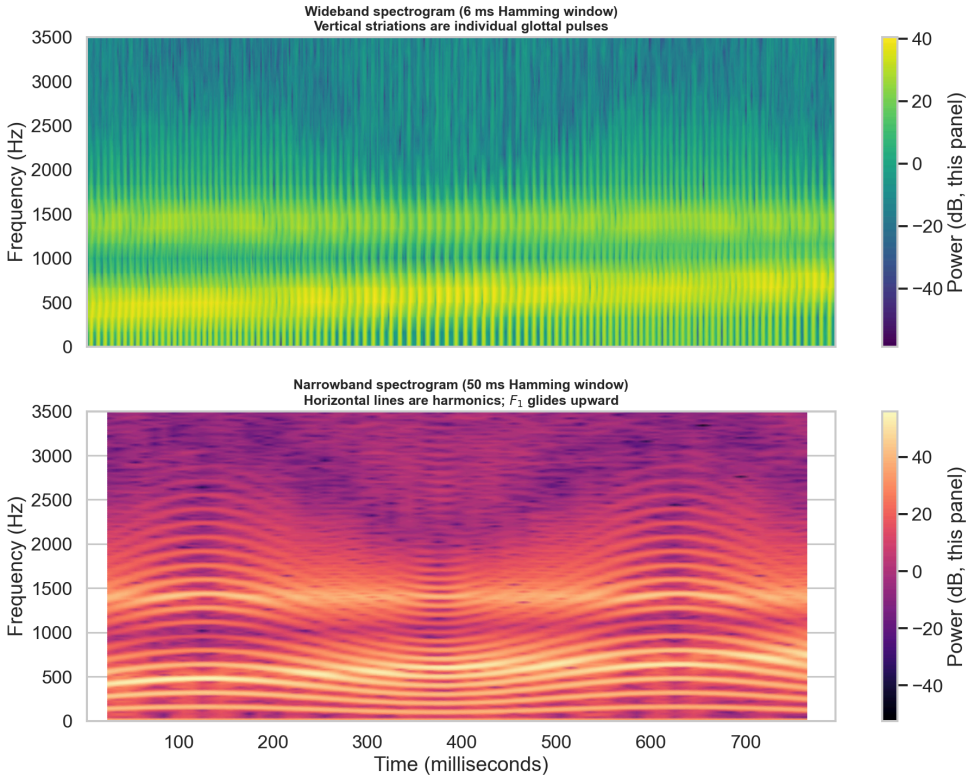


Figure 8. *Wideband vs. Narrowband Spectrogram comparison of a synthesized speech utterance. Top: Wideband spectrogram (6 ms window) resolves individual vertical pitch pulses. Bottom: Narrowband spectrogram (50 ms window) resolves individual horizontal harmonic frequency lines and fine formant bands.*

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7. The Source-Filter Deconvolution Paradigm & Cepstral Analysis

7.1 Gunnar Fant's Acoustic Model

Speech is modeled as the convolution of glottal excitation $e[n]$, vocal tract filter $v[n]$, and lip radiation $r[n]$:

s[n] = e[n] * v[n] * r[n] \quad \xrightarrow{\mathcal{F}} \quad S(\omega) = E(\omega) \cdot V(\omega) \cdot R(\omega)
Taking the logarithm converts multiplication into addition:

$$\log|S(\omega)| = \log|E(\omega)| + \log|V(\omega)|$$

Applying the Inverse Discrete Fourier Transform yields the **Real Cepstrum** $c[n]$:

$$c[n] = \mathrm{F}^{-1}\{\log|S(\omega)|\}$$

- **Low Quefrency ($|n| < 3$ ms):** Slowly varying vocal tract formant envelope $\log|V(\omega)|$.
- **High Quefrency ($n \approx T_0$):** Rapidly oscillating glottal excitation harmonics $\log|E(\omega)|$.

Applying a low-pass lifter ($c_{\text{liftered}}[n] = c[n]$ for $|n| \leq n_{\text{cutoff}}$) isolates the smoothed vocal tract formant envelope.

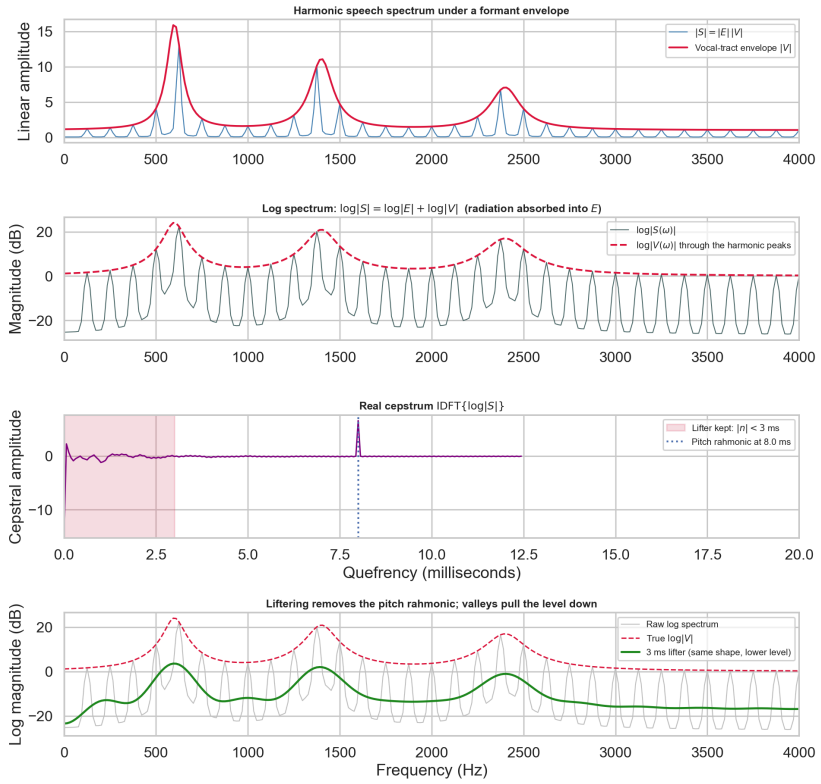


Figure 9. *Homomorphic cepstral deconvolution pipeline. 1: Speech spectrum exhibiting harmonic spikes under a resonant formant envelope. 2: Log spectrum where excitation and filter add linearly. 3: Real cepstrum on the quefrency axis showing low-quefrency formant coefficients and a distinct pitch spike at $T_0 = 8$ ms. 4: Liftered vocal tract formant envelope recovered by discarding high-quefrency pitch spikes.*

8. Psychoacoustic Mel Filter Banks & Feature Extraction

8.1 The Mel Scale Formula

The **Mel Scale** maps physical frequency f_{Hz} to perceived pitch f_{mel}:

$$f_{\text{mel}} = 1125 \ln\left(1 + \frac{f_{\text{Hz}}}{700}\right) = 2595 \log_{10}\left(1 + \frac{f_{\text{Hz}}}{700}\right)$$

Below 1000 Hz, the mapping is approximately linear; above 1000 Hz, it becomes logarithmic.

8.2 Triangular Mel Filter Bank Construction

Given M filters spanning $[f_{\text{low}}, f_{\text{high}}]$:

1. Convert endpoints to Mels: m_{\mathrm{low}} = \operatorname{mel}(f_{\mathrm{low}}), m_{\mathrm{high}} = \operatorname{mel}(f_{\mathrm{high}}).
2. Space $M+2$ points linearly in Mel space and convert back to Hertz f_i.
3. Map Hertz frequencies to DFT bin indices $k_i = \lfloor (N_{\text{fft}}+1) f_i / f_s \rfloor$.
4. Form triangular weighting filters $H_m[k]$.

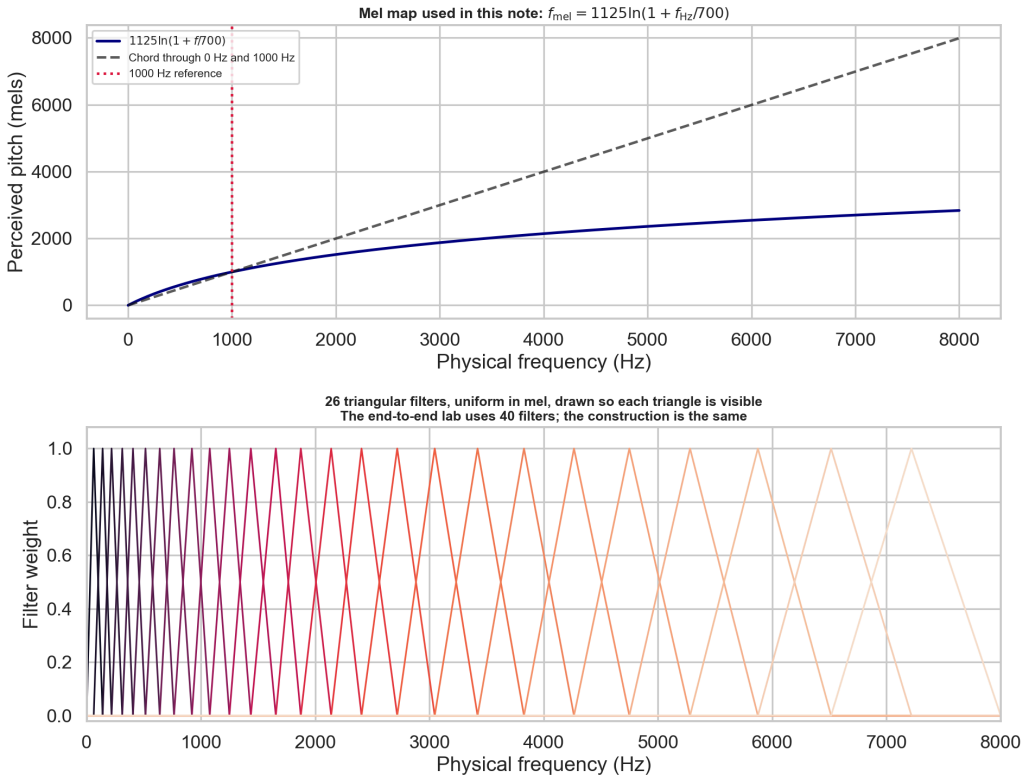


Figure 10. *The Mel scale and 26-channel triangular filter bank across* $[0, 8000]$ Hz. *Top: Mel transformation curve showing the transition from linear to logarithmic scaling around* 1000 Hz. *Bottom: Overlapping triangular filter bank with narrow, dense filters at low frequencies and broad, sparse filters at high frequencies.*

9. End-to-End Computational Pipeline

9.1 Runnable Speech Synthesis and Log-Mel Spectrogram Pipeline

```
import numpy as np

# Week 1 Master Pipeline: Synthetic Utterance to Log-Mel Spectrogram
fs = 16000
duration = 0.20  # 200 ms synthetic voiced speech
t = np.arange(int(fs * duration)) / fs

# Fundamental frequency contour (pitch glide from 120 Hz to 140 Hz)
f0 = np.linspace(120, 140, len(t))
phase = 2 * np.pi * np.cumsum(f0) / fs
excitation = (1.0 + np.sin(phase)) * (phase % (2 * np.pi) < 0.25)

# 2-pole vocal tract formant filter (F1 = 700 Hz, F2 = 1300 Hz)
r1, theta1 = 0.95, 2 * np.pi * 700 / fs
r2, theta2 = 0.92, 2 * np.pi * 1300 / fs
p1, p1_c = r1 * np.exp(1j * theta1), r1 * np.exp(-1j * theta1)
p2, p2_c = r2 * np.exp(1j * theta2), r2 * np.exp(-1j * theta2)

den = np.poly([p1, p1_c, p2, p2_c]).real
y = np.zeros(len(t))
for n in range(len(t)):
    feed = excitation[n]
    for k in range(1, len(den)):
        if n - k >= 0:
            feed -= den[k] * y[n - k]
    y[n] = feed

# STFT Framing (25 ms window, 10 ms hop)
frame_len = int(0.025 * fs)
hop_len = int(0.010 * fs)
n_frames = (len(y) - frame_len) // hop_len
n_fft = 512
win = np.hamming(frame_len)

stft_frames = []
for i in range(n_frames):
    segment = y[i * hop_len : i * hop_len + frame_len] * win
    spec = np.abs(np.fft.rfft(segment, n_fft))**2
    stft_frames.append(spec)
stft_mat = np.array(stft_frames).T

# Mel Filter Bank (40 channels)
n_mels = 40
def hz_to_mel(f): return 1125.0 * np.log(1.0 + f / 700.0)
def mel_to_hz(m): return 700.0 * (np.exp(m / 1125.0) - 1.0)

mel_pts = np.linspace(hz_to_mel(0), hz_to_mel(fs / 2), n_mels + 2)
hz_pts = mel_to_hz(mel_pts)
bin_pts = np.floor((n_fft + 1) * hz_pts / fs).astype(int)
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fbank = np.zeros((n_mels, n_fft // 2 + 1))
for m in range(1, n_mels + 1):
    for k in range(bin_pts[m - 1], bin_pts[m]):
        fbank[m - 1, k] = (k - bin_pts[m - 1]) / max(1, bin_pts[m] - bin_pts[m - 1])
    for k in range(bin_pts[m], bin_pts[m + 1]):
        fbank[m - 1, k] = (bin_pts[m + 1] - k) / max(1, bin_pts[m + 1] - bin_pts[m])

log_mel_spec = 10 * np.log10(np.dot(fbank, stft_mat) + 1e-6)

print(f"Synthesized Audio Duration: {duration*1000:.0f} ms | Samples: {len(y)}")
print(f"STFT Matrix Dimensions (Freq Bins x Frames): {stft_mat.shape[0]} x {stft_mat.shape[
print(f"Log-Mel Spectrogram Dimensions (Mel Bins x Frames): {log_mel_spec.shape[0]} x {log_
print(f"Dynamic Range: Min {np.min(log_mel_spec):.2f} dB, Max {np.max(log_mel_spec):.2f} dB
print(f"Filter 1 Center: {hz_pts[1]:.1f} Hz | Filter 40 Center: {hz_pts[40]:.1f} Hz")
```

9.2 Exact Terminal Output

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Synthesized Audio Duration: 200 ms | Samples: 3200
STFT Matrix Dimensions (Freq Bins x Frames): 257 x 17
Log-Mel Spectrogram Dimensions (Mel Bins x Frames): 40 x 17
Dynamic Range: Min -11.91 dB, Max 67.71 dB
Filter 1 Center: 44.4 Hz | Filter 40 Center: 7481.4 Hz
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10. Comparative Benchmarks & Practice Problems

10.1 Telecommunication Audio Standards

Audio Quality Standard	Bandwidth (f_{max})	Nominal f_s	Nyquist Threshold	Bit Depth (B)	Uncompressed Bitrate	Inc Sta
PSTN Telephony (Narrowband)	300\ \mathrm{Hz}\text{--}3.4\ \mathrm{kHz}	8 kHz	6.8 kHz	8-bit (μ-law/A-law)	64.0 kbps	ITU G.7 PC
HD Voice (Wideband)	50\ \mathrm{Hz}\text{--}7.0\ \mathrm{kHz}	16 kHz	14.0 kHz	16-bit linear PCM	256.0 kbps	AM / G
Studio Audio (Fullband)	20\ \mathrm{Hz}\text{--}20.0\ \mathrm{kHz}	44.1 kHz/48 kHz	40.0 kHz	16-bit / 24-bit PCM	1411.2 kbps (stereo)	CD Bro

10.2 Practice Problems with Complete Solutions

Problem 1 (Nyquist & Aliasing)

Question: A speech recording containing frequency content up to $7.2\ \text{kHz}$ is sampled at $12\ \text{kHz}$ without an anti-aliasing filter. What apparent frequency will a $7.0\ \text{kHz}$ harmonic fold into?
Solution:

$$f_{\text{apparent}} = |f - k \cdot f_s| = |7000 - 1 \times 12000| = |-5000| = 5000\ \text{Hz} = 5.0\ \text{kHz}$$

The $7.0\ \text{kHz}$ component aliases into $5.0\ \text{kHz}$, corrupting the vocal tract formant structure.

Problem 2 (Complex Pole Placement for Formant Filter)

Question: Construct the complex conjugate poles for a digital resonator placed at $F_1 = 800\ \text{Hz}$ with 3-dB bandwidth $B = 100\ \text{Hz}$ at $f_s = 16\ \text{kHz}$.
Solution:

$$\theta = \frac{2\pi F_1}{f_s} = \frac{2\pi(800)}{16000} = \frac{\pi}{10} = 0.3142\ \text{rad} = 18^\circ$$

$$r = \exp\left(-\frac{\pi B}{f_s}\right) = \exp\left(-\frac{\pi \cdot 100}{16000}\right) = \exp(-0.0196) \approx 0.9806$$

The pole pair is placed at $p_{1,2} = 0.9806 e^{\pm j 0.3142}$. Because $|p| = 0.9806 < 1$, the filter is BIBO stable.

Problem 3 (Mel Frequency Inversion)

Question: A Mel filter bank channel is centered at $m = 1200\ \text{Mels}$. Compute its center frequency in physical Hertz.
Solution:

$$f_{\text{Hz}} = 700\left[\exp\left(\frac{m}{1125}\right) - 1\right] = 700\left[\exp\left(\frac{1200}{1125}\right) - 1\right] = 700[\exp(1.0667) - 1] \approx 700[2.9057 - 1] = 1334.0\ \text{Hz}$$

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11. Common Pathologies & Engineering Traps

1. **Downsampling Without Low-Pass Filtering:**
   - Decimating audio from $48\ \text{kHz}$ to $16\ \text{kHz}$ by discarding samples without an anti-aliasing filter causes high frequencies to fold into the baseband, introducing severe audible distortion.
2. **Spectral Leakage from Rectangular Windows:**
   - Rectangular windowing has a slow $1/\omega$ sidelobe decay ($-13\ \text{dB}$ first sidelobe), masking low-energy formants adjacent to dominant fundamental frequencies. Tapered windows (Hamming/Hann) suppress sidelobes to $-43\ \text{dB}$.
3. **Pitch Doubling/Halving in Autocorrelation:**
   - Unconstrained peak picking picks formant cycles (pitch doubling) or subharmonic vocal fry pulses (pitch halving). Physiological limits (50\text{--}500\ \mathrm{Hz}) must constrain

search.

4. **IIR Filter Instability via Coefficient Quantization:**
   - Implementing high-order filters in direct form shifts denominator roots outside the unit circle ($|p_k| \geq 1$) under finite-precision rounding. Always decompose high-order filters into cascades of 2nd-order biquad sections.

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12. Academic References & Master Formula Sheet

References

1. **Rabiner, L. R., & Schafer, R. W. (2010).** *Theory and Applications of Digital Speech Processing*. Prentice Hall.
2. **Oppenheim, A. V., & Schafer, R. W. (2009).** *Discrete-Time Signal Processing (3rd ed.)*. Pearson.
3. **Fant, G. (1960).** *Acoustic Theory of Speech Production*. Mouton & Co., The Hague.
4. **Davis, S., & Mermelstein, P. (1980).** *Comparison of parametric representations for monosyllabic word recognition*. IEEE TASSP, 28(4), 357–366.

Master Formula Sheet

Concept	Formal Mathematical Expression	Dimensional Units
Nyquist Criterion	$f_s \geq 2f_{\max}$	Hz (Samples/s)
Quantization SQNR	$\text{SQNR} \approx 6.02B + 1.76$	dB
Discrete Convolution	$y[n] = \sum_k x[k]h[n-k]$	Amplitude
Forward DFT	$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$	Complex Amplitude
Frequency Resolution	$\Delta f = \frac{f_s}{N} = \frac{1}{T_{\text{record}}}$	Hz/bin
Autocorrelation	$r_{xx}[l] = \sum_n x[n]x[n-l]$	Energy (Amp_2)
Pitch Estimation	$F_0 = \frac{f_s}{l_*}, l_* = \text{argmax}_l\, r_{xx}[l]$	Hz
STFT	$X(m,\omega) = \sum_n x[n]w[n-mR]e^{-j\omega n}$	Complex Spectrum
Mel Scale	$f_{\text{mel}} = 1125 \ln(1 + \frac{f_{\text{Hz}}}{700})$	Mels
Real Cepstrum	$c[n] = \mathcal{F}^{-1}\left\{ \log	S(\omega)